



AMIT KUMAR



PROFESSIONAL SUMMARY

A conscientious **Social Architect** and **Data Scientist, Policy Designer** with a proven record of building high-impact data intelligence suborganisations from the ground up. I operate from a foundation of **first-principles-thinking**, blending deep, multi-disciplinary knowledge in social-technology, political philosophy, ethics, justice, science, and social systems to design and execute solutions to complex, large-scale challenges. My work is defined by a focusing on creating permanent, ethical, and self-sufficient state, public or civic-organizational assets that deliver strategic advantage and can serve as valuable resources to people at large. My approach is to not just build algorithms and databases, but to build people, state, public infrastructure, and institutional capacity.

As a systems thinker, I have also developed a comprehensive framework for **21st-century policy architecture**, conceptualizing scalable, social-technology-driven solutions for foundational challenges in governance, economic justice, and digital rights. I am driven to build humane, people-centric platforms that strengthen democratic institutions and create a more equitable society.

Alma Mater: **IIT KANPUR**

Kanpur, UP, India

(2009-2014*)

Integrated Master and Bachelor in Aerospace Engineering *

* **DROPPED OUT** during 10th (final) semester (this section reflects the status post 9th semester)

Note: Made a principled decision to depart before formal degree conferral to pursue a self-directed, interdisciplinary education better aligned with the mission of solving complex societal challenges.

Master's Thesis: "Design of Regional Airliner" Supervisor: Prof. A. K. Ghosh *

New Ideas Introduced: Distributed inflatable fuel storage mesh; In-flight aircraft CG variation for fuel efficiency; Using Smart Skin on wings & exposed surface for added aerodynamic variations

SELF STUDY Post Drop-Out

India

(2013-2019-now)

- Immersed self in a world of knowledge, reading over 600 Books & several online Courses, research papers, politics & policies, debates, labs & experimental works, new innovations, paradigms, social changes, talks, and multidisciplinary ideas from all domains to get a holistic & rigorous worldview
- Diagnosed several yet unrecognized maladies in Climate Change, Public Policy, Statecraft, Education, Gender, Equality, Architecture, Collaboration, Digital Platform, Environment, Labor
- Conceptualized solutions addressing them using concepts out of Systems Design, Law, Behavioural Economics, Sociology, Neuroscience, AI, Science, Game Theory, Politics, Networks, Democracy
- Notable Courses:** Reasoning, Data Analysis, and Writing, Art and Science of Economic Policy, Statistical Analysis of fMRI Data

WORK EXPERIENCE

Chief Intelligence Officer

Data Intelligence Unit, Indian National Congress (AICC)

(Aug'24- Aug'25)

Conceived, architected, and built the party's first permanent, in-house Data Intelligence Unit, establishing its strategic vision, technical infrastructure, operational workflows, and ethical framework. Operated as the unit's sole Tech Lead, Data Strategist, and Project Manager, transforming a nascent concept into the strategic nerve centre of the national organization. The work was pivotal in bridging the chasm between raw, complex data and clear, actionable political strategy, establishing a new paradigm for data-driven decision-making within the organization.

- **Institution Building & Leadership:** Built the DIU under severe budget and technical constraints, recruiting and transforming a team of novices with little to no technical background into a high-performing unit of data analysts and developers. Imparted foundational knowledge in computer science, data ethics, GIS mapping, and algorithmic thinking.
- **Electoral Roll Download and Digitization:** Built mechanisms and institutional capacity to download and digitise electoral rolls, in the fastest, cheapest and most accurate manner.
 - **Novel OCR mechanism:** For the complex first page of electoral rolls containing booth information, developed a groundbreaking **Optical Region Recognition (ORR)** approach, using image ensemble of image recognition methods, and precise region extraction, helping in finding regions containing specific text sections, which then were stitched back together with seamless precision. This bypassed traditional character-level OCR of whole page, reducing the cost of digitizing this specific page by an incredible **150 times** while guaranteeing 100% accuracy.
 - The overall process was **15 times faster** (~15 minutes per Assembly Constituency vs. the 5-hour industry standard) than leading data vendors, establishing a new benchmark for efficiency and **3-fold** cost-effectiveness.
 - Continuously refined processes using better and faster free libraries and self-built methods, doubling download speeds for the later states processed on a single laptop—giving up added cost due to earlier reliance on using multiple AWS machines for the states processed at the initial stage.
 - Built 3 CRNN models to solve the captchas that appear on ECI Eroll Website, further cutting down AntiCaptcha API costs. It was trained on around 10000 captcha images, with solutions consisting of 23 lowercase characters and 8 digits. This model results in correctly solving 100% of captchas.
 - Automated CAPTCHA Solving with Deep Learning: Developed and trained a suite of Convolutional Recurrent Neural Network (CRNN) models to autonomously solve complex CAPTCHAs from the ECI E-Roll website, eliminating reliance on costly third-party APIs.
 - **Training:** Built a training approach, feeding images to the model with minimal preprocessing: achieving ~100% validation accuracy without complex image-data cleaning, within 0.6 seconds (0.07 seconds per solve with batch processing utilising cached model loading), instead of using AntiCaptcha API, wasting ~8 seconds—and dollars.
 - **Model Architecture:** The model architecture combined a Convolutional Neural Network (CNN) for visual feature extraction with a Bidirectional LSTM to interpret the features as a character sequence, making it highly effective for this task. The model was trained on a supervised dataset of ~10,000 labelled CAPTCHA images (utilising AntiCaptcha solutions, which upon later analysis, were correct only about on ~86% occasions) and then was retrained only on correct ~8400 image-solutions pairs, achieving further accuracy increment.
 - Formalised methods to track and monitor monthly ERoll Additions, Deletions, Modification lists for Maharashtra, Delhi, Bihar, and generalised the process for entire country. Added mechanisms to check Minority Bias in ERoll modifications.

- **Cutting-Edge Community Inference Model:** Developed a model to infer community demographics from electors' data with unprecedented coverage. The model was able to cover **87.8% people** from Bihar, UP, MP (**95.3% with relaxed inference**), a significant leap from the previous 65% state-of-the-art benchmark, achieved by larger teams and organisations.
- **Party-Wide Mapping & GIS Capability:** Established and led all data-rich mapping tasks for the entire organization, continuously enhancing the GIS mapping capabilities. Curated and rectified inaccurate official shapefiles to produce vastly more precise and strategically valuable mapping assets.
- **Data Collection:** Built several tools to extract and scrape large datasets from several public websites, APIs, repositories. Implemented several wrangling strategies (multistep-complex-cleaning, sorting, etc) to store them in our structured repositories. Built streamlined python ETL Pipelines to access and share them as per the need. Created methodologies and Templates for collecting ground data from our ground workers, designed and used python NLP+ML workflow to analyse the text data, to understand the issues and pain-points of people in Bihar, Gujarat.
- **Gujarat District Profiles ETL:** Scrapped, wrangled and formalised all data resources from multiple resources such as MOSPI (microdata: National Family Health Survey, Periodic Labour Force Survey, Household Consumption Expenditure Survey, Time Use Survey), ECI, State and District websites, Election results (Assembly and Booth Level: past 6 elections), Booths, Grammanchitra, Bhuvan, and several other websites, and created full ETL to creating Wikipedia like district webpages, showcasing all data along with theoretical explanations grounded in real news and social media research (6 months prior to past 6 elections to 1 month after the elections).
- **Delivered a fully searchable, structured, and permanent data repository** to the party, ensuring sovereign ownership of critical intelligence assets and breaking the cycle of dependency on external vendors.
- **Data Based Strategy:** Formalised methods to develop pollical strategies –appointment, candidate selection, narrative building—based on data.
- **Evidence-Based Policy and Scientific Rigor:** In an era where political discourse is often dominated by populist rhetoric and ungrounded claims, the application of the scientific method—humility in the face of evidence, a willingness to be proven wrong, and a commitment to empirical validation— ensured that political strategy is grounded in reality, making the entire democratic enterprise more effective, more honest, and more resilient.

Senior Data Scientist

Indian National Congress

(Jan'24- July'24)

As a Lead Data Scientist for the General Election 2024, National War Room, laid the foundational data and technology infrastructure for the party's national campaign efforts. Built long-term, scalable, and consolidated data systems that continue to serve as core party assets. The work was approached not as a series of isolated tasks, but as an act of institution-building.

- **Foundational Data Architecture (General Election 2024):** Architected, conceived and delivered the party's first centralized, structured data repository. Integrated and standardized chaotic, disparate data streams from all national, state-level units, INC's frontal organisations, party leaders and workers, creating a permanent, searchable 'single source of truth'.

- **Organizational Standardization & Scalability:** Established a canonical system of unique, human-readable IDs and standardized spellings for all political and administrative units. This foundational work eliminated data ambiguity and enabled seamless, scalable data integration across the entire organization for the first time.
- **Automation of Strategic Intelligence:** Built and automated a system to generate and distribute constituency-wise 'Data Kits,' equipping all parliamentary candidates with actionable, localized intelligence on demographics, community breakdown, and electoral history. This system became the party standard and continues to be used for all subsequent assembly elections.
- **Localised Election Results Digitization and standardization:** Led end-to-end collection and digitization of Form 20 (booth-level results) data. Designed the process, built automated templates and digitisation pipelines, ensuring 100% accuracy and creating a permanent, structured repository of priceless democratic data. Trained and managed teams on data structuring principles, creating a permanent, digital archive for granular election analysis.
- **Geospatial Intelligence Capability:** Established the party's entire GIS mapping capability from scratch. Created standardized templates and developed detailed strategic maps integrating prior election results over Google Maps layer to visually analyse their constituencies.
- **Optimized Survey & Campaign Operations:** Developed systems for creating randomized and targeted datasets for survey calls (GOTV, Exit Polls) and tracking Guarantee Card and resource distribution, significantly reducing costs while enhancing the accuracy and auditability of campaign feedback loops.

Policy & Systems Design

Indian National Congress,
Independent Research, Self-
Publishing

(2018-current)

As a core part of my work, I have designed first-principles-based policy frameworks aimed at solving foundational societal challenges through innovative systems. Key proposals include:

- **Expanded Rights of People:** A framework for guaranteeing not just abstract rights but the physical, digital, and economic infrastructure required to make them a lived reality for every citizen, based on 21st century capabilities of humanity.
- **Reimagining Goods' and People's Movement:** A system design to transform the gig economy using a cooperative, cable-and-hub-based infrastructure, ensuring dignified work, public ownership, and efficient, sustainable delivery.
- **Open-Source Digital Platforms Ecosystem:** A blueprint for a suite of publicly built and owned, open-source digital platforms—from communication to e-commerce—designed to foster innovation, protect citizen data, and prevent monopolistic control.
- **Reconceptualised NYAY:** An advanced iteration of the Universal Basic Income concept, designed with robust, fraud-resistant delivery mechanisms and integrated with broader economic reforms to create a true social safety net.
- **Land Re-Acquisition and Re-Distribution:** A modern, data-driven approach to land reform that aims to correct historical inequities and optimize land use for sustainable and equitable development.
- **Aadhaar & UIDAI:** A critical analysis and proposed redesign of the national identity system to enhance privacy, reduce coercive exclusion, and ensure the system serves citizens rather than the state.

Coordinator

IT for Change

(Feb'23-Oct'23)

As Coordinator for the National Coalition on the Education Emergency (NCEE), played a key role in:

- **Proposal Development:** Drafted compelling grant, research, and funding proposals, and spearheaded applications to join influential global networks.
- **Advocacy:** Planned, executed, and disseminated campaigns to raise awareness and drive policy changes.
- **Research:** Supported research studies on education initiatives, ensuring widespread dissemination through press conferences, webinars, and social media.
- **Stakeholder Engagement:** Networked and mobilized diverse stakeholders, fostering collaboration and commitment to children's rights and education.

- **Event Management:** Organized impactful online and on-site events ([SIA/FRE](#), [Digital Edu-Conclave](#)), meetings, and consultations to advance NCEE's mission.
- **Communications:** Managed social media outreach, created newsletters, and maintained the website to enhance NCEE's visibility and impact.
- **Digital Literacy:** Led and supported workshops to empower teachers and students in government schools with essential digital skills.

Volunteer

Indian National Congress

(May'22-Now)

- Assisting party workers in several day-to-day party works with my tech, design, new ideas, research, campaign, ideation skills in their works, which aim to bring Sarvodaya and Antyodaya
- Collaborating with Sangam team in designing **Voices of Manipur Newsletter** after the start of conflict
- Helped Political Consultants with quick tech solutions, ideas, strategies during Karnataka Election

Political Strategist & Data Analyst

Indian National Congress, UP

(Oct'21-Mar'22)

- Within the *Ladki Hoon, Lad Sakti Hoon (I'm a Woman, therefore, can Fight too!)* campaign core team, helped strategize on ways to bring more women in politics
- Managed all tech and data needs, vis a vis. generating new reports, verifying and auditing all reports, D2D campaign Auditing, candidate selection & profiling, managing competing stakeholders, Auditing & Analysis of campaign collateral distribution, helping other teams create better data models
- **Program Lead:** Led and Managed the Campaign Ambassador Program, strived to attain at least 40% female participation (campaign goal) within the core campaign team, and to have their own voices

Community Leader, India Chapter

[RadicalxChange](#) Foundation, US

(Aug'21-Now)

- Voluntarily helping build partnerships with people and communities working on civic tech, digital governance and social sector, across India & Worldwide, in order to advance plural democracy ideas

Analyst

Grab.in, Gurugram

(Oct'20-Oct'21)

- Responsible for daily reports and day-to-day data analysis for the operations team
- After observing company's overall work & data structure, proposed new ideas & mechanisms, eg: digital literacy, truth seeking UX design: robust data, and better and easier conditions for riders

Volunteer Consultant

Indian National Congress

(Jan'19-May'19)

- Worked under Prof. [Rajeev Gowda](#) (ex-Chairman, Research Department, INC) in election manifesto and campaign policy proposal preparation and dissemination, and ideation on [several](#) issues
- [Proposed](#) novel, well researched, and rigorous implementation mechanism for the key election policy proposal on Minimum Income Guarantee Scheme (NYAY)

Volunteer

Aam Aadmi Party

(Sep'19-Dec'19)

Proposed innovative policy proposals for the AAP government, on a volunteer basis to Gopal Mohan (advisor to CM Arvind Kejriwal), a few of which got partially implemented

Lead Research and Culture Team Member of **Kanhaiya Kumar's** Team during Jana-Gana-Mana Yatra across Bihar in **protest** of CAA-NRC-NPR, Jan-Feb'2020

Teaching Assistant

Structures Lab, AE, IIT Kanpur

(July'13-Dec'13)

Assisted in conducting course, assignment solutions, grading processes, lab & experimental setups

Research Assistant

IIT Kanpur

(July'12-Dec'12)

Assisted: Subrahmanyam Saderla (PhD.) & Prof. A. K. Ghosh; **Project:** UAV Parameter Estimation

CORE COMPETENCIES

- **Systems Architecture & First-Principles Design:** Social Systems Design, First-Principles Policy Formulation, Algorithmic Thinking, Game Theory, Behavioural Science, Governance Models, Strategic Foresight.
- **Data Science & Engineering:**
 - **Languages:** Python, SQL, Stata
 - **Libraries:** Pandas, NumPy, Scikit-learn, SpaCy, TensorFlow, PyTorch, GeoPandas, GDAL, Rasterio, Google Earth Engine API, ArcGIS Rest API, geoai-py, libraries for parallel processing, NLP & Big Data.
 - **GIS & Geospatial Analysis:** QGIS, Geospatial Data Architecture, Shapefile Curation, Location-based Intelligence.
 - **LLM & Generative AI:** API Integration (ChatGPT, Gemini, Claude, OpenRouter, HuggingFace), Fine-Tuning for custom models, Grounding for factual accuracy, Structured Data Extraction.
- **Leadership & Strategy:** Institution Building, Team Leadership & Mentorship, Project Architecture, Data Strategy, Capacity Building, Ethical Frameworks.

SKILLS

Systems Thinking	Creativity, Innovation	Research, Analysis	Philosophy, Political
Design Thinking	Decision Making &	Digital Navigation	Science & Pol. Economy
Critical Thinking	Problem Solving	Digital World & Society	Policy Design & Analysis
Strategy Design	Creative & Academic	Sciences, Coops, Digital	Deep Observer
Idea Conceptualization	<u>Writing</u> , Documentation	Platforms, Governance	Intellectual Debates

AWARDS & ACHIEVEMENTS

- Secured All India Rank 92 in GATE'13 among ~15000 candidates
- Granted postgraduate **scholarship** by MHRD for the duration of June 2013-Jan 2014
- Percentiles of 99% in IIT-JEE 2009, AIEEE 2009, and 99.9 % in 2007 and 2009 board exams
- Received **Aviation Development Award and Scholarship** by SSPC, IIT Kanpur, 2011
- **Special Recognition Awardee** (Science & Technology Council, IITK), 2012: for several aeromodelling achievements, **pioneering** RC flight at IIT Kanpur, **mentorship** in aeromodelling activities & competitions, **cultivating** RC Flight & Aeromodelling as part of IIT Kanpur culture

MAIN PROJECTS/ TERM PAPERS/ PRESENTATIONS

Styrofoam Plane	(Summers Project)	(May'10-June'10)
Designed and fabricated first ever model to successfully fly in IITK — built from scratch in campus		
Conceptual Design of a Light Commuter Aircraft		(July'12-April'13)
(Annual Project: Design, Part Fabrication, Demonstration, Report & Presentation)		
Conceptualized an aircraft capable of flying 23 people for 800 km: selected as best design project		
3D Aerobatic Remote-Controlled Sports Aircraft	(Course Project)	(July'11-Nov'11)
Led a team of 7 members to best project award in aeromodelling course		
Computational General Equilibrium Modelling (CGEM)	(Course Project)	(May'11-June'11)

Studied the **macroeconomic response** of global economy upon sudden changes in economic factors, natural disasters, policies, technology or other external factors

Aerodynamics involved in Architectural Design

(Course Project)

(Jan'12-April'12)

EXTRA-CURRICULAR ACTIVITIES

- Secretary, Aeromodelling Club-IITK, 2010-11, Senior Mentor 2011-13
- Assistant Coordinator, Take-Off, Techkriti'11 (IITK annual technical festival)
- **Volunteer Educator:** Held supplementary classes encompassing interdisciplinary subjects using innovative examples from daily lives with live demos, attended enthusiastically by underprivileged children, under PRAYAS, a social initiative from IITK during 2010-2011